

$$1. \int \frac{(x^2-1)dx}{(x^4+3x^2+1)\arctan\left(\frac{x^2+1}{x}\right)}$$

$$\begin{aligned} * u &= \arctan\left(\frac{x^2+1}{x}\right) \\ du &= \frac{1}{1+\left(\frac{x^2+1}{x}\right)^2} \left(1-\frac{1}{x^2}\right) dx \\ du &= \frac{x^2-1}{x^4+3x^2+1} dx \end{aligned} \quad \left| \quad \int \frac{du}{u} = \ln|u| + C = \ln\left|\arctan\left(\frac{x^2+1}{x}\right)\right| + C \right.$$

$$2. \int (1 + \ln^2(x))^2 \left(\arctan(1 + \ln^2(x)) \right) \left(\frac{\ln(x)}{x} \right) dx$$

$$\begin{aligned} 1 + \ln^2 x &= u \\ \frac{2 \ln x}{x} dx &= du \\ \arctan u &= m \quad u^2 du = dv \\ \frac{du}{u^2+1} &= dm \quad \frac{u^3}{3} = v \end{aligned} \quad \left| \quad \int \frac{1}{2} u^2 \arctan u du = \frac{1}{2} \left(\frac{u^3}{3} \arctan u - \int \frac{u^3}{u^2+1} du \right) \right.$$

$$\begin{aligned} \int \frac{u^3 du}{u^2+1} &= \frac{1}{2} \int \frac{u^2 (2u du)}{u^2+1} = \frac{u^2}{2} - \frac{\ln(u^2+1)}{2} \\ I &= \frac{u^3}{6} \arctan u - \frac{u^2}{12} + \frac{\ln(u^2+1)}{12} + C = \frac{(1+\ln^2 x)^3}{6} \arctan(1+\ln^2 x) - \frac{(1+\ln^2 x)^2}{12} + \frac{\ln(1+\ln^2 x+1)}{12} + C \end{aligned}$$

$$3. \int \frac{e^{e^{\ln(\cos(2x))} - \ln(\sec(2x))} + e^{\cos(2x) + 4 \ln(x^{1/4} \sec(2x))}}{e^{\cos(2x) - 3 \ln(\csc(2x))}} dx$$

$$\begin{aligned} \int \frac{e^{\cos 2x} \cdot \cos 2x + e^{\cos 2x} \cdot 2 \sec^4 2x dx}{e^{\cos 2x} \cdot \sec 2x} &= \int \frac{\cos 2x dx}{\sec 2x} + \int \frac{2 \sec^3 2x dx}{\sec 2x} \\ &= -\frac{1}{4} \cos^2 2x - \frac{\cos 2x}{2} + \frac{\sec 2x}{4} + C \end{aligned}$$

$$4. \int \frac{e^{\ln(\ln(x))^2 + \ln(x) + 2}}{x(\ln(x))^3 + 8x} dx$$

$$\begin{aligned} \int \frac{\ln^2(x) + \ln x + 2}{\ln x + 8} \frac{dx}{x} &= \int \frac{u^2 + u + 2}{(u+2)(u^2-2u+4)} du = \frac{1}{3} \int \frac{du}{u+2} + \frac{1}{3} \int \frac{2u+1}{u^2-2u+4} du = \frac{1}{3} \int \frac{du}{u+2} + \frac{1}{3} \int \frac{2u-2}{u^2-2u+4} + \int \frac{du}{(u+1)^2+3} \\ \ln x &= u \\ \frac{dx}{x} &= du \end{aligned} \quad \begin{aligned} \frac{u^2+u+2}{(u+2)(u^2-2u+4)} &= \frac{A}{u+2} + \frac{Bu+C}{u^2-2u+4} \\ u^2+u+2 &= u^2A-2uA+4A+Bu^2+Cu+2Bu+2C \\ u^2+u+2 &= u^2(A+B)+u(-2A+C+2B)+4A+2C \\ \begin{cases} A+B=1 \\ -2A+C+2B=1 \\ 4A+2C=2 \end{cases} &\Rightarrow \begin{cases} A=\frac{1}{3} \\ B=\frac{2}{3} \\ C=\frac{1}{3} \end{cases} \end{aligned} \quad = \frac{1}{3} \ln|\ln x| + \frac{1}{3} \ln|\ln x - 2 \ln x + 4| + \frac{2}{3} \arctan\left(\frac{\ln x + 1}{\sqrt{3}}\right) + C$$

$$5. \int \frac{dx}{x(\ln(x)-1)^3(\ln(x)+2)^4}$$

$$\text{Integral de Chia: } \int \frac{dx}{(x+a)^m(x+b)^n} = \frac{1}{(b-a)^{m+n-1}} \int \frac{(t-b)^{m+n-2}}{t^m} dt$$

haciendo $t = \frac{x+a}{x+b}$

$$I = \int \frac{d(\ln x)}{(\ln x - 1)^3 (\ln x + 2)^4} = \frac{1}{3^6} \int \frac{(1-t)^5}{t^5} dt = \frac{1}{3^6} \int \left(\frac{1}{t^5} - \frac{5}{t^4} + \frac{10}{t^3} - \frac{10}{t^2} + 5t - t^5 \right) dt$$

$$\text{haciendo } t = \frac{\ln x - 1}{\ln x + 2} \quad (1-t)^5 = 1 - 5t + 10t^2 - 10t^3 + 5t^4 - t^5$$

$$I = \frac{1}{3^6} \left[-\frac{(\ln x - 1)^2}{2} + 5 \left(\frac{\ln x - 1}{\ln x + 2} \right) + 10 \ln \left| \frac{\ln x - 1}{\ln x + 2} \right| - 10 \left(\frac{\ln x - 1}{\ln x + 2} \right) + \frac{5}{2} \left(\frac{\ln x - 1}{\ln x + 2} \right)^2 - \frac{1}{3} \left(\frac{\ln x - 1}{\ln x + 2} \right)^3 \right] + C$$

$$6. \int \frac{(x^4+4x^3+11x^2+12x+8)dx}{(x^2+2x+3)^2(x+1)}$$

$$\begin{aligned} \int \frac{(x^4+4x^3+11x^2+12x+8)}{(x^4+4x^3+10x^2+12x+9)(x+1)} dx &= \int \frac{dx}{x+1} + \int \frac{x^2-1}{(x^2+2x+3)^2(x+1)} dx = \int \frac{dx}{x} + \frac{1}{2} \int \frac{2x+2}{(x^2+2x+3)^2} dx \\ &= \ln|x| - \frac{1}{2} \ln(x^2+2x+3) + C \end{aligned}$$

$$7. \int \frac{(3+11\cos(x))dx}{\cos(x)(1+3\cos(x))}$$

$$\cos x = u \rightarrow \frac{3+11u}{u(3u+1)} = \frac{3}{u} + \frac{2}{3u+1}$$

$$\rightarrow \int \frac{3+11\cos x}{\cos x(1+3\cos x)} dx = 3 \int \frac{dx}{\cos x} + 2 \int \frac{dx}{3\cos x+1} = 3 \ln|\sec x + \tan x| + \frac{1}{4\sqrt{2}} \ln \left| \frac{\tan x + \sqrt{2}}{\tan x - \sqrt{2}} \right| + C$$

$$I_1 = \int \frac{dx}{3\cos x+1} = \frac{1}{2} \int \frac{dt}{3(1+\frac{3}{2}t)+1} = \frac{1}{2} \int \frac{\sec^2 x dx}{4-2t \tan x} = \frac{1}{4} \int \frac{d(\tan x)}{2-t \tan x} = \frac{1}{8\sqrt{2}} \ln \left| \frac{\tan x + \sqrt{2}}{\tan x - \sqrt{2}} \right|$$

$$x=2t$$

$$dx=2dt$$